

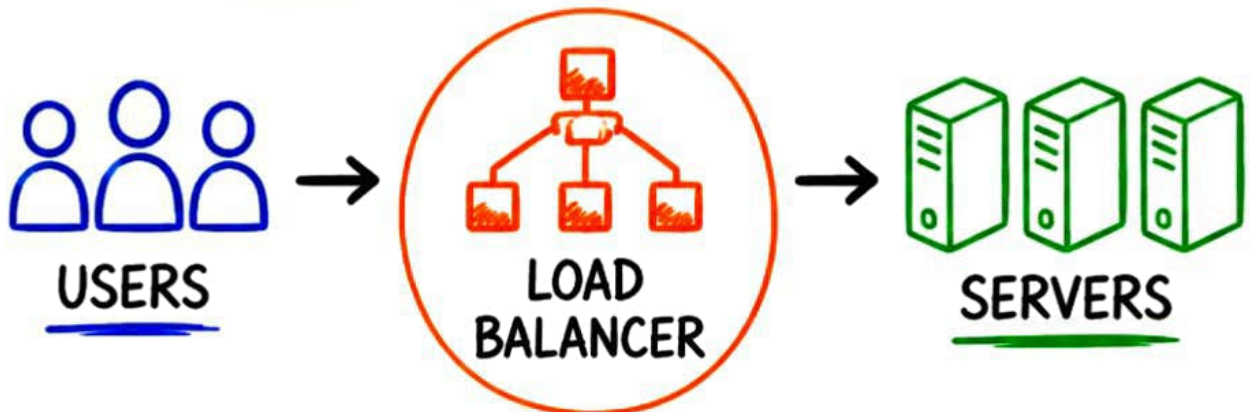


@NeerajSingh-DevOps

AWS LOAD BALANCER

COMPLETE NOTES

BEGINNER → ADVANCED



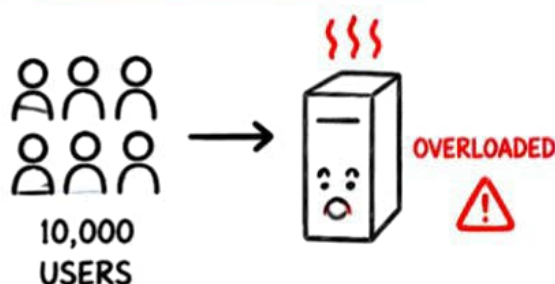
DISTRIBUTE • SCALE • STAY AVAILABLE

★ One Load Balancer. Many Healthy Servers.

AWS ELASTIC LOAD BALANCING
MADE EASY

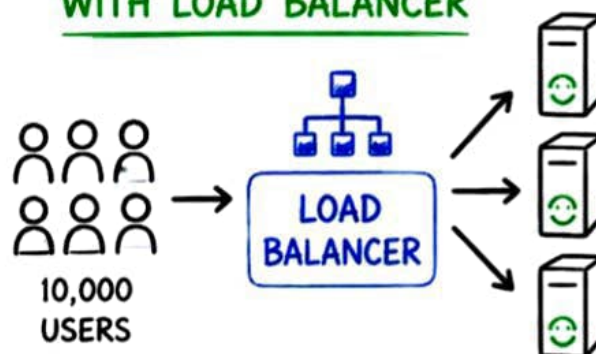
② WHY DO WE NEED A LOAD BALANCER?

WITHOUT LOAD BALANCER



High load on single server
→ Slow or Failure!

WITH LOAD BALANCER



Traffic is distributed across healthy servers
→ High Availability!



DEFINITION

A Load Balancer automatically distributes incoming traffic across multiple healthy targets to improve availability, scalability and performance.

WHY IT'S IMPORTANT?


HIGH AVAILABILITY

Healthy servers keep serving traffic.


SCALABILITY

Handles more traffic by adding more servers.


BETTER PERFORMANCE

Distributes load evenly.


SECURITY

Hides backend servers from direct access.


COST EFFECTIVE

Optimizes resource utilization.



MEMORY LINE: ONE ENTRY POINT → MANY SERVERS

③ TYPES OF AWS LOAD BALANCERS

① ALB

Application Load Balancer

LAYER 7



Works with
HTTP / HTTPS

KEY FEATURES

- ✓ Path-based routing
- ✓ Host-based routing
- ✓ HTTP headers
- ✓ WebSocket support
- ✓ Redirects & rewrites



BEST FOR

Web applications,
APIs, Microservices

EXAMPLE



example.com/orders
→ Orders Service

② NLB

Network Load Balancer

LAYER 4



Works with
TCP / UDP / TLS

KEY FEATURES

- ✓ High performance
- ✓ Ultra low latency
- ✓ Handles millions of requests/sec
- ✓ Static IP support
- ✓ TLS pass-through



BEST FOR

High performance
apps, TCP/UDP
workloads, Gaming,
IoT, Finance apps

EXAMPLE



TCP : 3306
Database traffic

③ GWLB

Gateway Load Balancer

LAYER 3



Works with
GENEVE Protocol

KEY FEATURES

- ✓ Deploy & scale network appliances
- ✓ Transparent traffic inspection
- ✓ Centralized insertion point



BEST FOR

Security appliances,
Firewalls, IDS/IPS,
Traffic inspection

EXAMPLE



Traffic → Firewall Appliance
→ Your VPC



EASY MEMORY TRICK

ALB = Application
(Layer 7, HTTP)

NLB = Network
(Layer 4, TCP/UDP)

GWLB = Gateway for
Security Appliances

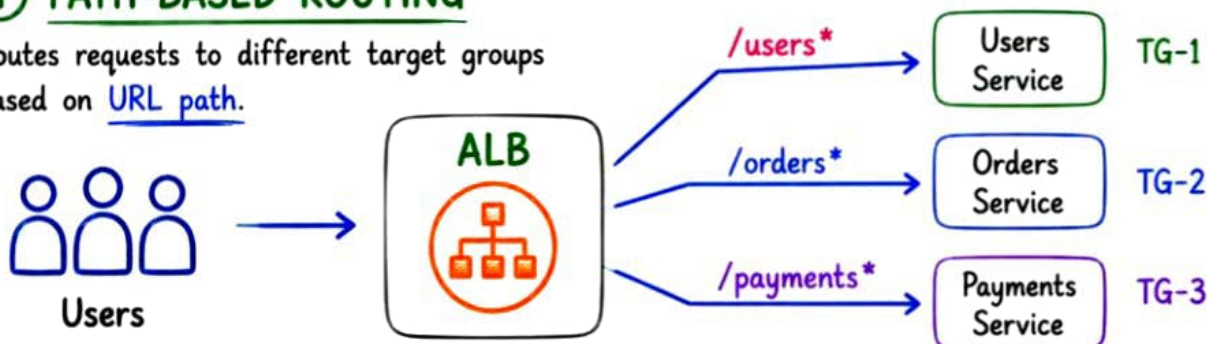


④ HOW ALB ROUTES TRAFFIC ≡

ALB is a smart **Layer 7** load balancer.

① PATH-BASED ROUTING

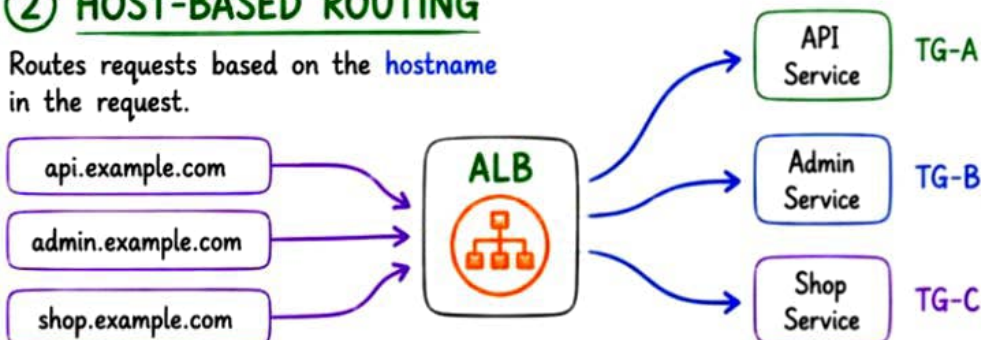
Routes requests to different target groups based on URL path.



 **Example:**
example.com/users → Users Service
example.com/orders → Orders Service
example.com/payments → Payments Service

② HOST-BASED ROUTING

Routes requests based on the hostname in the request.



 **Example:**
api.example.com → API Service
admin.example.com → Admin Service
shop.example.com → Shop Service

KEY TAKEAWAY

- ★ ALB understands HTTP/HTTPS requests.
- ★ It can route traffic using **paths** or **hostnames**.
- ★ Great for microservices and multi-app environments.

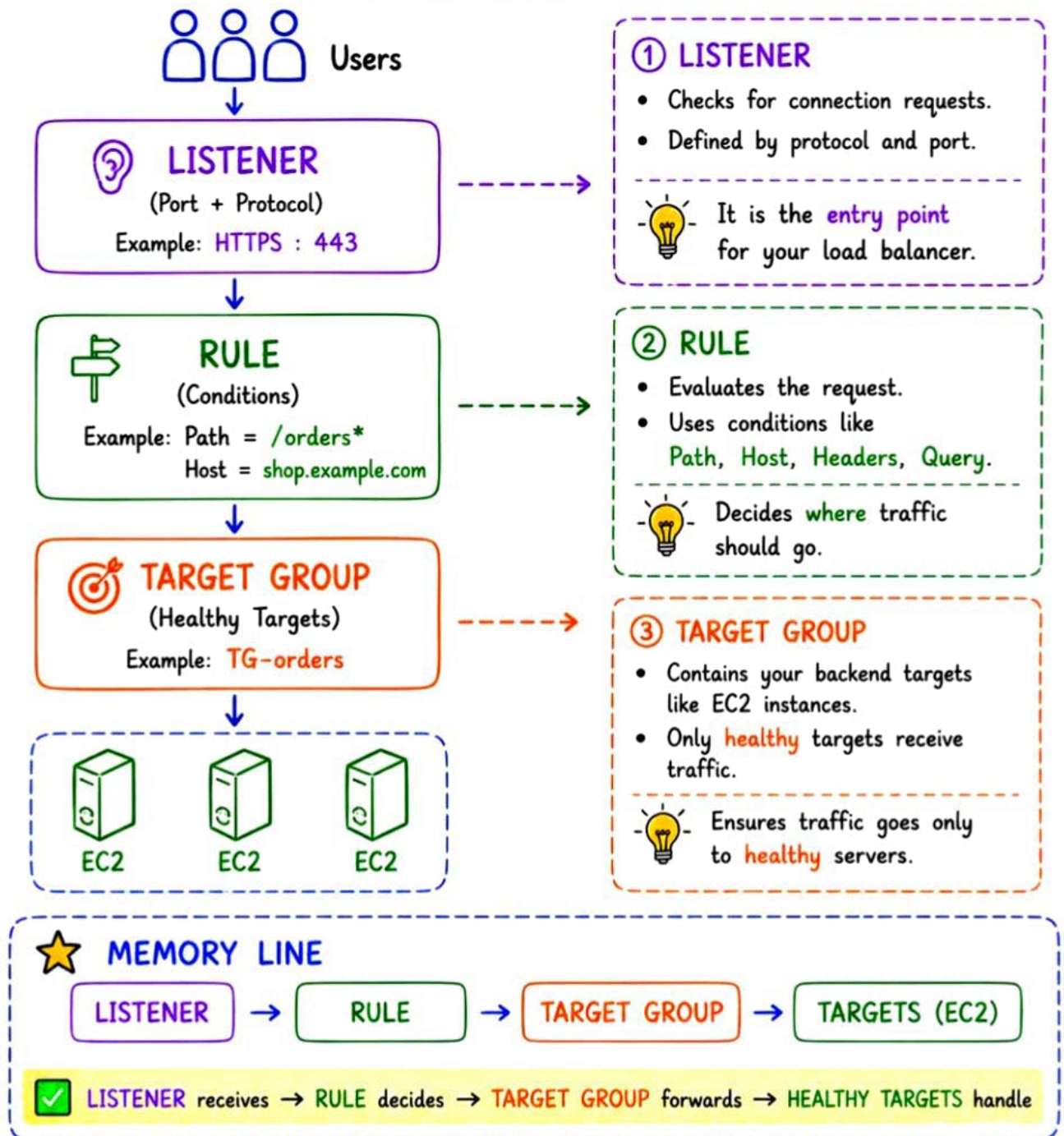


MEMORY TRICK:

ALB looks at the **application** (Layer 7) and sends the request to the **right destination**.

5 LISTENER + TARGET GROUP

How traffic flows from User to EC2



TIP: One Load Balancer can have multiple Listeners, Rules and Target Groups.

⑥ ≡ LOAD BALANCER ≡

Distributes incoming traffic across multiple targets to improve availability, scalability and performance.

WHAT IS IT?



A Load Balancer is a traffic distribution service that routes requests to multiple servers (targets) based on defined rules.

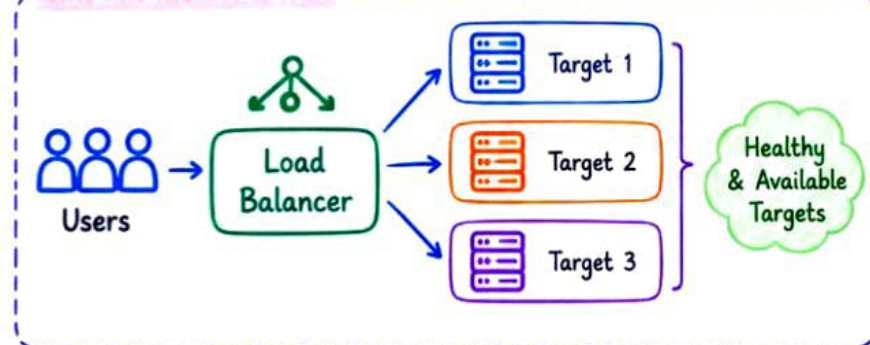
HOW IT WORKS

- 1 Receives client requests.
- 2 Checks target health.
- 3 Routes traffic to healthy targets using rules (e.g., round robin, least connections).
- 4 Monitors and removes unhealthy targets.

TYPES

- 1 **Application Load Balancer** (Layer 7 - HTTP/HTTPS)
 Best for web apps and microservices.
- 2 **Network Load Balancer** (Layer 4 - TCP/UDP)
 Best for high performance and low latency.
- 3 **Gateway Load Balancer** (Layer 3)
 Best for security appliances and firewalls.

ARCHITECTURE



KEY FEATURES

- ★ High availability
- ★ Auto scaling
- ★ Health checks
- ★ SSL/TLS termination
- ★ Sticky sessions (optional)
- ★ Cross-zone support

REAL WORLD EXAMPLE



MEMORY TRICK:

Load Balancer = Load distributes, so no single server gets all the stress!

⑦ = ALB DEEP DIVE =

Application Load Balancer = **Smart traffic distribution** for modern apps.

OSI MODEL

ALB operates at

LAYER 7

(Application Layer)

7
6
5
4
3
2
1

KEY BENEFITS

- ✓ Intelligent routing (path / host / headers / query)
- ✓ High **availability** across AZs
- ✓ Auto scaling
- ✓ Integrated with AWS services
- ✓ Improves performance and reliability



LISTENERS



- Define the protocol and port (HTTP : 80, HTTPS : 443)
- Entry point for traffic

RULES



- Evaluate incoming requests
- Match conditions
- Forward to the appropriate target group

TARGET GROUPS



- Group of targets (EC2, IP, Lambda)
- Health checks to ensure only **healthy** targets receive traffic
- Supports multiple target types

HEALTH CHECKS



- Periodically check target health
- Unhealthy targets are removed
- Improves application reliability

ARCHITECTURE OVERVIEW

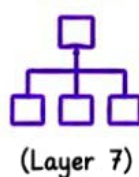


Users



Internet

ALB



(Layer 7)

Rule 1

/orders*

TG - Orders



EC2 Instances

Rule 2

/payments*

TG - Payments



EC2 Instances

Rule 3

/users*

TG - Users



EC2 Instances



COMMON USE CASES

- ✓ Web applications
- ✓ Microservices
- ✓ Multi-tenant applications
- ✓ API traffic management

PRO TIP



Use ALB for **layer 7 routing** (content-based).
Use NLB for **layer 4 routing** (performance).



MEMORY TRICK:

Think **ALB** = **A**pplication **S**mart **R**outer

It listens, decides and delivers to the right target!

⑧ ALB COMPONENTS EXPLAINED

Understanding each component helps you design better and troubleshoot faster.



1 LISTENER

- Listens for incoming client requests.
- Defined by protocol and port.
- Example: HTTP : 80
HTTPS : 443



2 RULE

- Evaluates the request based on conditions.
- Conditions can be path, host, headers, query string, etc.
- Each rule has a priority.



3 TARGET GROUP

- Group of targets (EC2, IP, Lambda) that receive traffic.
- Performs health checks.
- Only healthy targets get traffic.



4 TARGET

- Actual destination, e.g., EC2 instance, IP address, or Lambda.
- Must be registered in a target group.
- Health status determines if it receives traffic.



5 HEALTH CHECK

- ALB checks target health periodically.
- Configurable: protocol, path, interval, timeout, thresholds.
- Unhealthy targets are removed automatically.



6 SSL/TLS CERTIFICATE

- Used by HTTPS listeners.
- Decrypts traffic at ALB.
- ACM certificate (recommended) or IAM certificate.

HOW THESE COMPONENTS WORK TOGETHER



Users



Request comes in → Listener receives → Rule evaluates → Target Group selects healthy targets → Request is forwarded → Response is returned to the user.



LISTENER PORTS

Port	Protocol
80	HTTP
443	HTTPS
Others	Custom (if needed)

COMMON RULE CONDITIONS

- ✓ Path: /api/, /images/
- ✓ Host: app.example.com
- ✓ Headers: User-Agent, Content-Type
- ✓ Query String: id=123
- ✓ Source IP (via WAF integration)



MEMORY TRICK:

L - Listen → R - Rules → TG - Target Group → T - Targets
ALB makes smart decisions and sends traffic to the right place!

9 ALB vs NLB vs CLB

Choose the **right load balancer** for the right use case!

FEATURE	ALB Application Load Balancer	NLB Network Load Balancer	CLB Classic Load Balancer
 Layer	Layer 7 (Application)	Layer 4 (Transport)	Layer 4 (Transport)
 Protocol	HTTP, HTTPS, WebSocket, gRPC	TCP, UDP, TLS	TCP, SSL/TLS, HTTP, HTTPS
 Use Case	Web applications, Microservices, APIs	High performance, Low latency apps, Gaming, IoT	Legacy applications
 Performance	Good	Very High	Good
 IP Type	DNS name	Static IP (Elastic IP)	DNS name
 Health Check	Advanced (Path, Host, HTTP Codes)	TCP, HTTP, HTTPS, TLS	TCP, HTTP, HTTPS, SSL
 Cross-Zone Load Balancing	Enabled by Default	Enabled by Default	Enabled by Default
 Sticky Sessions	Yes (Cookie based)	Yes (Source IP)	Yes (Cookie based)
 WAF Integration	Yes	No	No
 Pricing	Per LCU	Per NLB Hour + LCU	Per Hour

WHEN TO USE?



Use ALB when:

- You have HTTP/HTTPS traffic
- Need content-based routing
- Working with microservices
- Need WAF + advanced routing rules



Use NLB when:

- You need ultra high performance
- Handling TCP/UDP traffic
- Gaming, IoT, Streaming apps
- Static IP is required



Use CLB when:

- You have a legacy application
- Simple TCP/HTTP routing
- Migrating from EC2-Classic



PRO TIP: ALB for smart routing, NLB for speed, CLB for legacy!



MEMORY TRICK: ALB = Application (Smart) | NLB = Network (Speed) | CLB = Classic (Legacy)

10 BEST PRACTICES FOR ALB

Follow these best practices to build **secure**, **scalable** and **cost-effective** ALB.



① Use HTTPS listeners and redirect HTTP to HTTPS.



② Enable cross-zone load balancing for even traffic distribution.



③ Use target group health checks with proper thresholds.



④ Use sticky sessions only when your application requires it.



⑤ Keep your ALB and target groups in the same VPC.



⑥ Use WAF to protect against common web exploits.



⑦ Monitor with CloudWatch and set alarms.

COMMON MISTAKES

- ✗ Using HTTP instead of HTTPS.
- ✗ Not configuring health checks properly.
- ✗ Registering unhealthy instances.
- ✗ Ignoring idle timeout settings.
- ✗ Not enabling access logs.
- ✗ Placing ALB in a public subnet without security.



IMPORTANT CLOUDWATCH METRICS



- RequestCount
- TargetResponseTime
- HTTPCode_ELB_5XX_Count
- HTTPCode_Target_5XX_Count
- HealthyHostCount
- UnHealthyHostCount
- ConsumedLCUs

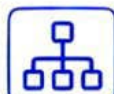
DESIGN CHECKLIST



Users



Internet



ALB



Target Group



Targets



Monitoring

✓ Identify users & traffic needs

✓ Ensure DNS is configured

✓ Create ALB with secure listeners

✓ Configure TG with proper health checks

✓ Register healthy targets

✓ Enable monitoring & logs

KNOWLEDGE RECAP

ALB

→ Intelligent routing

→ Multiple targets

→ High availability

→ Better performance

→ Happy users!



MEMORY TRICK:

A Smart Balance = High Availability + Speed + Security

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